

Paper XI · The ARC Theory · Law I, the ARC Principle · Convergent Evidence for Recursive Amplification: a Graded Cross-Domain Register

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The ARC Theory · OSF osf.io/dc9gw · every claim checkable

A register bearing on Law I and, as context only, on Law V: each row graded for the part of the principle it bears on, no total claimed, the pattern recorded as context, not confirmation.

*A dated convergence record, graded row by row against the canonical evidence register.
No numerical total is claimed. One structural principle. The closed loop.*

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ABSTRACT

Between 8 December 2024 and 1 September 2026, a 21-month window, research programmes, corporations, governments, and institutions independently arrived at the same structural principle: recursive self-correction produces capability gains exceeding linear accumulation, external alignment constraints are structurally insufficient to contain it, and the corrective must be embedded, not imposed. This convergence register is exploratory (an observational count, not a test statistic); the underlying structural principle that above-threshold recursive self-correction produces stably super-linear gains is on trial in draft study-k (the boundary) and draft study-ad (the correction exponent), both drafted, preregistered by timestamp (first anchored on 12 August 2026 at block 962066) and awaiting human submission. Two exceptions are disclosed. The Hernández-Espinosa to Gumbau chain is graded in the register as one dependent lineage, and the interfaith-governance rows belong to a single institutional movement older than the anchor. Apart from those two, no group cites another, and no group cites the author's work. The pattern is discovered, not manufactured. No numerical convergence total is claimed from these rows: the canonical graded register (33 rows, as amended 7 September 2026, under the 2 August 2026 policy) publishes no total, because technical experiments, theorems, engineering proposals, policy developments, cultural echoes, prior work and author-run studies cannot legitimately be summed as one evidential quantity. This paper is the convergence report. It documents each convergence with exact sources, dates, and gap measurements. It situates them within the ARC/Eden framework (Papers I-X). It closes the loop connecting the researcher's neurodivergent cognition (Paper C, PNP), an independent, peer-reviewed paper that argues for neurodivergent influenceability among AI agents as a contingent response to misalignment (Hernández-Espinosa et al., 2026, PNAS Nexus), and a preprint that credits that paper and states two impossibility results on the structural unverifiability of AGI alignment (Gumbau Mezquita, 2026, arXiv:2606.28639). The method is the hypothesis. The PNAS paper is about machine agents, not about me; the human parallel is argued separately, in Paper C, on its own terms. The chain of dates is independently checkable. The SHA-256 hash of the foundational manuscript is published. The sealed .eml is held privately and available on request; its SHA-256 is published, and the anchor establishes that those specific bytes existed no later than the anchored time, checkable by anyone against the public chain. Authorship, key custody, semantic content and absence of prior disclosure are separate questions carrying their own evidence, and are stated as such.

Scope note (19 September 2026). An owner review of this opening finds that it states more than the graded register is built to carry, and this note scopes it to what the rows establish. The same paragraph declares the register's own taxonomy: technical experiments, theorems, engineering proposals, policy developments, cultural echoes, prior work and author-run studies cannot legitimately be summed as one evidential quantity, and no total is published. Rows that are not commensurable enough to be summed are not commensurable enough to have reached the same three-part principle between them. The rows bear on different components of it: a mathematics result on unverifiability addresses whether an external constraint can be verified, and says nothing about the rate at which capability grows with depth or about where a corrective must sit; a policy development or a cultural echo addresses neither the growth rate nor the sufficiency of external constraint. Each row is accordingly graded on its own, for the component it bears on, and neither any single row nor the set as a whole is offered as establishing the conjunction of the three. Unchanged: the window and its dates; every row, its class and its source; the no-total policy; the two exceptions this paper names, the Hernández-Espinosa to Gumbau chain and the interfaith-governance rows; and the statement that no group cites another and no group cites the author's work, which is the documentary form.

Reading key

This paper carries a RESULT: the graded convergence register, in which each independently sourced observation is classified and evidence-graded rather than summed into a single number. Within The ARC Theory it sits as the empirical record against which the laws are checked, with the class taxonomy, the retained pre-grading snapshot and the four disclosed dependence groups reported line-by-line. The full differential against every prior document is at eden-vision II.A.8.

Status note (19 September 2026). An owner review of this description finds that it assigns the register a role that §6 of this paper states it has not yet earned, and this note aligns the two. §6 sets three conditions on any aggregate reading of these rows: independence, a disconfirming search with its denominator, and a base-rate comparison; and it states plainly that absent those, convergent evidence is indistinguishable from confirmation bias. Those conditions are recorded there as outstanding. This register is therefore a graded contextual record, published row by row with each row's class, source and dependence disclosure, and its aggregate reading is conditional on the §6 tests. The formulation this evidence supports is the one §1 already prints: the pattern warrants investigation as a candidate structural principle, and §6 sets the tests that would decide. Unchanged: the class taxonomy, the retained pre-grading snapshot, the four disclosed dependence groups, and §6 in full.

THE DEPENDENT CHAIN, STATED PLAINLY

A manuscript dated 8 December 2024 recorded the control-failure thesis and the embedded-correction direction. Sixteen months later a peer-reviewed paper argued, about AI agents, that perfect alignment is unattainable and engineered cognitive diversity can protect; a preprint crediting that paper then derived formal unverifiability results. One lineage, one direction, three dated documents. The register grades this as context, not confirmation, and this paper claims nothing stronger.

1. The Graded Convergence Record

The canonical home for this evidence is the graded register at /research/convergences/ (33 rows, generated 6 September 2026 and last amended 7 September 2026, under the claims-governance policy of 2 August 2026). **No numerical convergence total is published there or here.** The former summed register was assembled retro-

spectively, without a pre-outcome scoring rule, a fixed horizon, a base-rate model or a complete failure register, and its rows cannot legitimately be summed as one evidential quantity. The graded register instead assigns each row one primary evidence class: at the amendment of 7 September 2026, 1 formal, 2 qualified-technical, 2 engineering, 4 institutional, 6 prior-work, 12 context and 6 excluded, with the strongest rows graded moderate and the engineering rows weak. Four dependence groups are disclosed there. Two of them are the exceptions this paper names: the Hernández-Espinosa to Gumbau formal-limits chain is one dependent lineage, not two confirmations, and the interfaith-governance rows belong to one institutional movement that predates the December 2024 anchor. The other two are a compute-governance group, covering chips, fabrication, export control and hardware enforcement, which the register records as belonging to one policy and engineering literature that substantially predates the December 2024 anchor, and a group of rows that quote the staff of one organisation, which the register records as one organisational source whose rows are not independent of each other. Independent work belongs to its authors: a later event can show that a question matters without establishing this programme’s priority, a successful prediction or the truth of its answer.

Superseded snapshot, retained as this paper’s dated record (2 July 2026). The table below is the register as this paper first published it, before the 2 August 2026 grading policy. Its per-row classes (PREDICTION, CONVERGENT, CONCURRENT) and its gap arithmetic are superseded by the graded register above. Rows it graded CONVERGENT now grade variously prior-work, context, institutional, engineering or excluded; its row 2 in particular (Willow) is graded prior-work, because the below-threshold result was public as a preprint on 24 August 2024, before the 8 December 2024 manuscript, with the Nature version following on 9 December 2024. The author’s non-awareness of that preprint when writing the December 2024 manuscript is declared on the canonical row as biographical attestation; the grade does not move on it, because the public order is unchanged by awareness, and holding that line against the author’s interest is part of what makes the register’s favourable rows worth believing. Margins in the Gap column stated against arXiv submission dates (rows 1, 8, 9, 18, 19) carry an anchor taken from arXiv’s record alone (arXiv Atom), and first public disclosure has not been separately verified for those rows.

#	Domain	Convergence	Eastwood artefact date	External date	Gap (anchor: artefact date)	Class	Primary source
1	AI Safety	Anthropic alignment faking: 78% in the RL-training condition (12% baseline)	8 Dec 2024	18 Dec 2024	10 days	PREDICTION	arXiv:2412.14093
2	Quantum Computing	Willow: first below-threshold quantum error correction	8 Dec 2024	9 Dec 2024 (Nature; the preprint reporting below-threshold operation had been public since 24 Aug 2024)	1 day if the Nature paper is the marker; the anchor itself falls later than the preprint	CONVERGENT	Nature (2024), Google Quantum AI
3	AI Reasoning	The Sequential Edge: sequential beats parallel in 95.6% of configurations	8 Dec 2024 (one-sided directional statement); comparison formalised Jan 2026	4 Nov 2025	Concurrent: they precede the formalisation; priority theirs	CONCURRENT	arXiv:2511.02309
4	Classical Physics	Classical time crystal via nonreciprocal forces	Dec 2024 / papers Jan 2026	22 Apr 2025 (arXiv:2504.15495 v1; published in PRL 6 Feb 2026)	135 days after A1	CONVERGENT	PRL 136, 057201

5	Acoustics	First acoustic topological space-time crystals	Paper III (Jan 2026)	Feb 2026	Concurrent with Paper III	CONVERGENT	Newton 2, 100334 (Cell Press) [a commentary on Tong, Zhang, Li, Zhang and Qiu, Acoustic realization of monoatomic topological space-time crystals, Newton 2(2), 100304 ; the DOI of article 100304 was first registered on 18 November 2025]
6	Acoustics	First dispersive phononic time crystals (band folding, subharmonics)	Prediction code Mar 2026	Jun 2026	~3 months after prediction code	PREDICTION	Nature Communications
7	Hardware	Hardware-level ethics enforcement patents (runtime ethics gate)	30 Apr 2025 (A2)	Jan 2026 (filed Jul 2025)	3 to 9 months after A2	CONVERGENT	US 2026/0010411 + US 2026/0010780
8	Hardware	Embedded off-switches for AI compute: deadman-switch security anchors	30 Apr 2025	9 Sep 2025	~4.5 months after A2	CONVERGENT	arXiv:2509.07637
9	Hardware	FlexHEG tamper-proof guarantee processors (ARIA-commissioned)	30 Apr 2025 (A2)	Jun 2025 (arXiv:2506.15093; no earlier ARIA release date has been verified for this row)	on the arXiv date, a gap of ~1.5 months from A2	CONCURRENT	arXiv:2506.15093
10	Regulation	US orders Fable 5 + Mythos 5 offline: first export-control action against a deployed frontier model	2 Jan 2026 (Chokepoint)	12 Jun 2026	~5 months after book	CONVERGENT	Harvard Law Review blog; CSA whitepaper; Reuters (Jun 2026)
11	Geopolitics	Macron at G7 Evian: 'trusted partners' compute sovereignty framing	2 Jan 2026 (Chokepoint)	19 Jun 2026	~5.5 months after book	CONVERGENT	Broadband Breakfast; CNBC G7 coverage
12	Legislation	EU Tech Sovereignty Package (Chips Act 2.0 + CADA)	2 Jan 2026 (Chokepoint)	3 Jun 2026	~5 months after book	CONVERGENT	CNBC, 3 Jun 2026 (Virkkunen)

13	Religion	Pope Leo XIV first encyclical: Babel versus Jerusalem framing for AI	30 Apr 2025 (Babylon/Eden, A2)	25 May 2026	~13 months after A2	CONVERGENT	vatican.va; Church Times; NY Post (May 2026)
14	Religion / AI	Christopher Olah (Anthropic co-founder) co-presents encyclical at the Vatican	8 Dec 2024 (interfaith call, A1)	25 May 2026	~17 months after A1	CONVERGENT	NY Post, 25 May 2026; Church Times
15	Media / AI	Jack Clark (Anthropic co-founder): 'gas pedal but no brake pedal'	8 Dec 2024 (A1)	5 Jun and 19 Jun 2026	~18 months after A1	CONVERGENT	BBC World Service w3ct98k8; Euronews
16	Venture	Recursive Superintelligence: \$500M at \$4B for closed-loop recursive self-improvement; convergent naming with A1's core concept	8 Dec 2024 (A1)	17 Apr 2026	~16 months after A1	CONVERGENT	Sifted; TechFundingNews (Apr 2026)
17	Semiconductors	OpenAI + Broadcom Jalapeño inference chip (9-month tape-out, 10 GW target)	2 Jan 2026 (Chokepoint)	24 Jun 2026	~6 months after book	CONVERGENT	openai.com announcement; CNBC
18	Mathematics / Cognition	Neurodivergent influenceability as contingent solution to alignment	8 Dec 2024 (A1)	14 Apr 2026 (preprint May 2025)	~16 months after A1	CONVERGENT	PNAS Nexus pgag076 + arXiv:2505.02581
19	Mathematics	The Undecidability of AGI Alignment: Unverifiability Theorem + Trilemma	8 Dec 2024 (A1)	26 Jun 2026	~18.5 months after A1	CONVERGENT	arXiv:2606.28639

Row 3, stated at full strength and no further. Publication priority on the quantified finding, sequential beating parallel self-consistency in 95.6 per cent of configurations at matched compute, belongs to Sharma and Chopra (arXiv:2511.02309, 4 November 2025), credited unreservedly; the programme's compute-matched formalisation followed in Paper II (22 January 2026), designed and run before sight of their paper. The December 2024 manuscript carries a one-sided directional statement and an equation in which parallel width never had a term, so a parallel exponent near zero is a structural consequence rather than a surprise; that derivation was completed in 2026 by the operational definition of R , and the conclusion carries the date of its youngest premise. The two results are never equated: theirs measures chain refinement against parallel voting on reasoning benchmarks, Paper II measures error reduction in production chains; convergent direction, different instruments.

Register discipline: one external event per row, one primary source per row, hyperlinked to the official source wherever a stable canonical URL exists (arXiv, DOI, publisher, patent office, broadcaster, vendor); press-sourced rows name their outlets and date instead. Gaps are measured from the named artefact date, stated per row. Candidate rows are held back and are

neither counted nor shown until each carries a verifiable primary source, and the register as published carries no candidate list; reader engagement with the work (for example gifted copies) is kept out of the register and never counted as convergence. Priority is credited outward wherever the dates demand it (row 3). The full register: michaeldariuseastwood.com/research/convergences/. The evidence spine at michaeldariuseastwood.com/research/evidence-spine.html carries the verification model, the evidence rungs, the anchor cards and the append-only correction record, and publishes no quarantine or engagement list.

A dated, graded record: every row carries its primary source, its evidence class and its dependence disclosure. Independence is a tested property here, not an assumption: the register discloses its dependent chains, and §6 states the audits that would defeat the rest. The pattern warrants investigation as a candidate structural principle; §6 sets the tests that would decide.

2. Three Degrees of Neurodivergence

Degree 1 - The Researcher (Paper C, PNP). Michael Darius Eastwood. Diagnosed ADHD. Diagnosed autism. Adulthood diagnosis. Brain that connects domains uniform thinkers don't see as adjacent. Builds a 221,236-word framework across domains while representing himself in complex legal proceedings. The PNP (Polymathic Neurodivergent Profile) provides the descriptive clinical framework. Clinical documentation exists and is held privately. The method IS the hypothesis.

Degree 2 - The Protection Argument (Hernández-Espinosa et al., 2026, PNAS Nexus). Oxford, King's College London, Alan Turing Institute, University of Tokyo. The authors offer a mathematical proof, from formal undecidability and irreducibility arguments, that misalignment is inevitable and that full orchestrated controllability of agentic systems is impossible. Solution: cognitive diversity. Neurodivergent thinking is not a defect. It is the protective architecture. The paper does not cite Eastwood. Does not know he exists. Independently, and about machine agents, its argument is that cognitive diversity is the protective architecture; the parallel with humans is Paper C's to draw, not theirs.

Degree 3 - The Formal Proof (Gumbau Mezquita, 2026, arXiv:2606.28639). University Jaume I, Spain. Reads Hernández-Espinosa. Extends with Trakhtenbrot's Wall. Two formal theorems, the Unverifiability Theorem of Alignment and the Theorem of Finite Structural Unverifiability of AGI Alignment, with the Soundness-Completeness-Tractability Trilemma derived as their corollary. Credits Hernández as conceptual foundation. Does not cite this work. The direction is convergent, and the register grades the pair as one dependent chain: context, not confirmation.

Neither outside group had any knowledge of the author's work. Because the Gumbau paper is built on Hernández-Espinosa's, the register treats the two as a single lineage. None planned this. The chain is discovered, not manufactured.

3. The Closed Loop

The closed loop has four structural properties:

1. The method IS the hypothesis. The cognitive architecture that produced the framework - neurodivergent pattern recognition across domains - rhymes with what Hernández-Espinosa et al. argue for machine agents: that diversity of reasoning styles protects an ecosystem. The parallel is suggestive, argued in Paper C, and no more than that. The researcher is not a person who happens to have a theory. The researcher IS the theory's first evidence. The brain behind it exhibits the same type of reasoning diversity that the PNAS paper argues the field needs among machine agents.

2. The framework IS its own evidence. The register's rows record programmes arriving at the same structural conclusion without knowing about each other or the framework. This is not a claim of causation. It is an observation of convergence. The pattern holds across AI safety, quantum computing, classical physics, semiconductor manufacturing, Vatican doctrine, UN governance, venture capital, and mathematical logic. A pattern that appears across domains this different is discovered, not manufactured; how much it confirms is a graded question, row by row.

Evidence-class note (19 September 2026). *An owner review of the first two properties finds that they classify as evidence two things that are not measured outcomes of the proposed dynamics, and this note states what they are. How a framework came to be developed, and the cognitive architecture of the person who developed it, are intellectual history and motivation: they can explain why a question was asked and how a pattern was noticed, and they are not a measurement of whether recursive self-correction produces the gains the framework predicts. That the register's rows record programmes arriving at a similar structural conclusion is an observation about the literature, graded row by row, and not a demonstration by the framework of itself. The qualifiers already printed beside these sentences are the governing ones: the parallel is suggestive, argued in Paper C, and no more than that; this is not a claim of causation; and the chain is context, not confirmation. Unchanged: both properties as stated claims about method and pattern; Paper C's separate argument; the domain list; and the graded treatment of every row.*

3. The proof IS independent of the researcher. Gumbau Mezquita's formal theorems do not depend on believing Eastwood. They do not depend on Eastwood's methods. They do not depend on Eastwood's sources. Derived from Trakhtenbrot's Wall and Finite Model Theory, credited to Hernández-Espinosa and published on arXiv on 26 June 2026, they are formal mathematical results. Either they hold or they do not; as published they stand, and nothing has refuted them at the time of writing. And they say what Eastwood said 18 months earlier.

Comparison note (19 September 2026). *An owner review of this comparison finds that it sets an earlier directional statement beside later formal theorems derived, as this same paragraph records, from a distinct formal foundation, and this note applies the standard §1 already prints for exactly this kind of comparison. That standard is that two results are never equated where the instruments differ: convergent direction, different instruments. Read to that standard, the relationship here is a thematic one between a directional statement and a formal result reached by other means, and it is not an identity of content unless a crosswalk of premises and conclusions supports more. Unchanged: the dates; the credit to the external authors, which is given unreservedly and stays that way; the observation that the theorems stand as published; and the register's grading of this chain as one dependent lineage and as context rather than confirmation.*

4. Each link of the chain is independently verifiable. manuscript → later formal results in the same direction → the graded register that refuses to overstate them → Paper C stating the human framework on its own terms. Every date is independently verifiable. Every source is published. Every SHA-256 hash matches. The closed loop is not an argument. It is a chain of evidence.

4. Method

Convergences were identified through systematic search of 2025-2026 AI safety, quantum computing, hardware, semiconductor policy, Vatican doctrine, governance, neuroscience, and mathematics literature. Each convergence was tested against: (a) independent origin (no shared citations, personnel, or funding with Eastwood or with each other), (b) structural match (same principle, different substrate), and (c) temporal gap (date of Eastwood's articulation vs. date of independent confirmation). The Dec 8, 2024 manuscript was verified at line level against source files. All .eml files were SHA-256 verified. All external DOIs and arXiv IDs were confirmed.

5. Related work

The rival instrument

The work that sits closest to this programme's question, and so the paper this one should be weighed against, is Engels, J., Baek, D., Kantamneni, S. and Tegmark, M., "Scaling Laws For Scalable Oversight", arXiv:2504.18530, first posted 25 April 2025 at 17:54:27 UTC (a time taken from arXiv's record alone, arXiv Atom; a NeurIPS 2025 Spotlight Poster, confirmed on 15 August 2026 against the NeurIPS virtual page for poster 115536, as recorded in the statement paper's reference dossier). It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step.

The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. Their framework contains no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. Nested Scalable Oversight is iterated same-class oversight by construction, and this programme's central prediction is that the same-class ladder is bounded however many rungs are added, while a cross-class corrector is not. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have.

Antecedents and near-misses

On the question. Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", arXiv, 28 February 2012, read at its source), separating "speed from intelligence explosion" and undertaking to "consider possible bounds on intelligence", augmenting Chalmers' 2010 analysis. The question and the speed-versus-structure distinction are therefore at least fourteen years old. What that literature does not contain is a number: no measurable exponent, no derived ceiling, no architecture dependence.

On impossibility. Three 2025 arXiv papers argue that perfect control is unattainable: Yao, "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, v1 12 June 2025 02:30:30 UTC, a time taken from arXiv's record alone, independently observed by the Internet Archive on 13 June 2025; cited by arXiv:2512.03048); Yao, "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, the only paper in the arXiv abstract corpus containing the phrase "irreducible uncontrollability", abstract-search total of one, measured 12 August 2026); and Ball, Gluch, Goldwasser, Kreuter, Reingold and Rothblum, "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The third, notably, concludes that alignment "must instead be integrated into the model's architecture and weights", an independent argument, from filtering intractability, in the same direction as this programme's architecture dependence; it is convergent support on that leg, not a rival. Two of the three papers are by one sole author; the description "a wave" overstates the literature's breadth, though not the seriousness of the six-author paper.

On the mechanism. That anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates). The mechanism is not the claim. The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent.

The nearest structural analogue. The quantum error-correction threshold theorem also converts a qualitative worry into a critical value. It concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant; the analogy is one of method. This analogue was identified by the programme's own search rather than by a referee, and is disclosed accordingly.

On feedback and stability (classical). The general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage (small-gain theorems). The differential: those are gain conditions on interconnected systems, not a power-law exponent criterion on a corrector's scaling with the capability of a recursively improving system. Cited so the general proposition is never claimed; the exponent formulation, dynamics and estimator are the contribution surface.

The nearest quantitative neighbour. Liu, A. and Meng, J., "Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs", arXiv:2604.22273 (a single source, the abstract read via the paper's own html page, 12 August 2026), recasts self-correction as a closed-loop control problem via a two-state Markov model parameterised by an Error Introduction Rate and an Error Correction Rate, and derives a directly measurable stability threshold: iterate only when ECR/EIR exceeds $\text{Acc}/(1 - \text{Acc})$. The differential, stated wherever it is cited: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. No scaling exponent on the corrector, no capability-growth exponent, no reciprocal identity, no architecture or composition-class term. Complementary regimes; neither contains the other. Two adjacent 2026 empirical results from the same search, each to be verified at source before being cited beyond its title: arXiv:2601.00828 reports an accuracy-correction paradox, weaker models achieving materially higher intrinsic correction rates than stronger ones, per-step evidence directionally adjacent to the same-class cap and never to be quoted as an exponent measurement; and arXiv:2507.02778 (Self-Correction Bench) reports a systematic self-correction blind spot across open models.

On recursive creation. Smolin's cosmological natural selection is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously ("Echoing Smolin's cosmological natural selection, AI could create recursive universes with their own laws", read at its source in the author's notes).

On the field around the register. The questions this register tracks from outside now have a named home: Recursive Dynamics, the field the author founded, set out in its founding paper (v3.4 of 5 October 2026, DOI 10.17605/OSF.IO/HCPBU), whose objection A12 argues from within what this register documents from without: that the questions form one family no existing field claims whole. The register and the field's founding paper are one discipline pointed in two directions; the register grades what arrived from outside, and the founding paper states what would kill the name.

Correction note (3 October 2026). *Until this version, the paragraph above described Recursive Dynamics as proposed as a field in its founding paper, and called that paper the field proposal. It now names Recursive Dynamics as the field the author founded, and the paper as the field's founding paper. The epistemic-status line beneath the citation block said that what this programme names Laws are conjectures under registered adversarial test. Laws IV and V, added to the theory on 24 September 2026, are neither registered as laws nor tested, and the line now gives their standing apart from that of Laws I to III. In §2, Degree 2 said that Gödel/Turing prove external alignment is impossible. In its own abstract, the PNAS paper offers a proof that misalignment is inevitable and that full orchestrated controllability of agentic systems is impossible, and Degree 2 now says that. In the references, the entries for Paper C, for Infinite Architects, for the PNAS paper and for the Gumbau paper printed short forms of their titles. They now print the titles in full as their registries give them, the Gumbau paper's as the title of its first version. The entry for the manuscript of 8 December 2024 now says that it is held privately and has no online source. The abstract called the PNAS paper the independent paper proving that such cognition is a peer-reviewed argument for engineered cognitive diversity among AI agents, and called the Gumbau paper formal unverifiability results in a preprint. It now describes each in the terms of its own abstract. In §2, Degree 2 was headed The Protection Proof: the PNAS paper offers a proof that misalignment is inevitable, and argues for diversity among agents, so the heading now says The Protection Argument. Degree 3 named the Gumbau paper's two theorems as Unverifiability of Alignment and the Trilemma, and now names them as its first version does, with the Trilemma as their corollary. The abstract, §6 and §8 said of their draft registrations only that they await human submission; each is now also given as preregistered by timestamp, with*

the date of its first anchor. §5 and the references cited the founding paper at version 2.8; they now cite version 3.1 of 3 October 2026, in which objection A12 is the same objection. §1 and §5 printed working labels in capitals for how a source was read or dated, and §5 called the author's notes the operator's notes; both are now in plain words. In the retained snapshot, row 5 printed the identifier of a commentary without a link; the identifier is now linked, and a bracket names the research article that the commentary reports on and gives the day that article's DOI was first registered, 18 November 2025. That day is earlier than both dates the row prints, the date of the external work and the date of Paper III. Eighteen works that the text cites and the reference list lacked now have entries, each from its record; six of them are that commentary, that research article and the four works that rows 7 to 9 of the snapshot cite. The entry for Pope Leo XIV's encyclical gave only 25 May 2026, the day it was presented; it now also gives 15 May 2026, the day it was signed and the date the Holy See's own page gives it. No result, date or limit changes.

6. Falsifiability

This claim is defeasible, and the criteria below state exactly what would defeat it. The strength of convergent evidence rests entirely on the convergences being genuinely independent, so these tests are adversarial by design.

Independence failure. Each convergence must arise from a source causally independent of the author's manuscripts and of every other convergence (see §4). The claim collapses to selection bias if a re-audit finds that a material fraction of the convergences share a common origin, cite one another, or were produced by parties with documented prior knowledge of the 8 December 2024 manuscript.

Defeat threshold (count). The count is itself a falsifiable claim. On a re-audit conducted under a protocol drafted, preregistered by timestamp (unit 58, first anchored on 12 August 2026 at block 962066) and awaiting human submission, applying strict one-event-per-row deduplication and primary-source verification, the pattern is defeated if the number of surviving independent convergences falls below the level distinguishable from chance co-occurrence given the breadth of domains searched. The retained snapshot's nineteen rows were its verified rows alone. Because the graded register claims no total, the defeat test now bears on the independence of the graded rows rather than on any number, and candidates still under verification remain excluded until they are sourced.

Missing denominator (disconfirming search). As evidence, the claim is defeated unless an equally systematic search for *dis*-confirming cases (domains where added recursive depth does not produce super-linear-above-threshold gains) is conducted and reported. Without the denominator (domains searched versus domains matching), the convergence rate cannot be shown to exceed chance.

Look-elsewhere effect. Across as many distinct domains as are surveyed here, some coincidental structural similarity is expected. The claim is defeated if the observed convergence rate does not exceed the false-positive rate expected from pattern-matching across that many independent domains. **Rate-language note (19 September 2026).** *An owner review of these two tests finds that the narrowing recorded three paragraphs above has not been carried through to their wording, and this note carries it through. That narrowing states that because the graded register claims no total, the defeat test bears on the independence of the graded rows rather than on any number. A rate requires a numerator and a denominator, and with no total claimed there is no convergence rate here to compare against chance or against a false-positive expectation. Both tests are therefore read as tests on the rows: the first, that without a reported disconfirming search the register cannot be shown to be more than a selected sample of the domains read; the second, that across domains this varied some structural similarity is expected, so a row's structural match carries weight only where its grading and its dependence disclosure survive scrutiny. Restoring a rate would require first declaring a population, inclusion rules and a base-rate model, fixed in advance; none is declared, and none is proposed here. Unchanged: both tests as defeat conditions; the reverse-causation test and the engagement-versus-convergence rule; the statement that absent these tests convergent evidence is indistinguishable from confirmation bias; and the retained snapshot's record of its own verified rows.*

Reverse causation. A convergence counts only if the external result was not, directly or indirectly, influenced by the author's own dissemination. Any convergence traceable to the book, the website, or public posts rather than

pre-dating or arising independently of them is reclassified as engagement, not independent confirmation, and removed from the count.

Predictive failure. The structural principle makes forward predictions (that future recursive-depth results will show the same above-threshold super-linearity). Forward predictions drafted and dated as draft registrations awaiting human submission that fail at a rate inconsistent with the claimed universality defeat the structural reading, leaving only a set of historical coincidences.

What would *not* defeat the claim: the reclassification of any single convergence, or one source proving weaker than described: the claim is about the aggregate pattern, not any individual row. But the aggregate must survive the independence, denominator, and base-rate tests above. Absent those, "convergent evidence" is indistinguishable from confirmation bias, and this paper says so plainly.

7. What This Means

For the ARC/Eden programme. The framework is not speculating in a vacuum: the register's rows record independent workers reaching the same structural conclusions, and the corroboration is not from allies, collaborators, or people who know about the work; it is from people who do not know the work exists. How much weight that carries is a graded question, row by row, and no single number is claimed for it; the weight depends entirely on the rows being genuinely independent, which §6 sets out how to test.

Scope note (19 September 2026). *An owner review of this summary finds that it carries forward the aggregate reading the abstract has now scoped, and this note applies the same scoping here. The rows bear on different components of the structural principle and are graded individually; neither any row nor the set is offered as establishing the conjunction of all three components; and the aggregate reading remains conditional on the §6 tests. The sentence's own closing clause already says the weight is graded row by row and that no single number is claimed for it, and that is the reading. Unchanged: every row and its grading, the no-total policy, and §6.*

For the researcher. The PNP framework describes the cognitive architecture that built this. The PNAS paper argues that diversity of reasoning styles protects an ecosystem of machine agents. Whether the human parallel holds is exactly what Paper C puts on the table, as a stated framework rather than borrowed proof. The clinical documentation provides the medical record. The closed loop is complete. The researcher's claim to expertise is not institutional. It is evidential. The evidence record is published. The SHA-256 hashes match. The sealed .eml sources are available on request, hash-anchored.

For AI safety. Multiple independent programmes now argue that external alignment constraints are structurally insufficient, and several arrive at embedded correction that co-scales with capability as the direction, before the first decoupled recursive system ships. That pattern is context, not confirmation; the deciding tests are the drafted registrations (§6). The question is whether any institution will act on the structural argument before the evidence becomes a post-mortem.

This paper and Laws IV and V, 3 October 2026. *The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. Their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper is a dated convergence record: its convergences were identified by search (§4), and each row of its register is graded on its own (§1). Its abstract gives, as the third part of the structural principle it reports, that the corrective must be embedded, not imposed, and §7 records that several programmes arrive at embedded correction as the direction, a pattern it calls context, not confirmation. In the theory registration, embedded correction is carried by P21. P21 is one of the two propositions that carry Law V, and so this paper bears on Law V as context only. It compares no correction that takes part in each revision round with correction that*

sees only the finished output, varies no build order, and measures no gain installed by external post-training under later capability-directed training. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes no result printed here: the results, their dates and their limits stand as they were.

8. Framework predictions the convergence pattern points to

The eclipse. The framework's sharpest disagreement with current practice concerns what oversight is made of. A corrector built from the same substrate as the system it corrects cannot anti-correlate with its own errors, so its correction exponent is bounded above, under the independence-accumulation model, by one half and in practice sits below it; a corrector from a different composition class carries no such bound. The scalable-oversight programme, weak-to-strong generalisation, debate, amplification, recursive reward modelling and constitutional methods, is built overwhelmingly from same-class correctors, and its unstated premise is that this scales. This framework predicts it is capped. The law and the value can be pulled apart. What owns the ground is the law: a self-improving system stays correctable only for as long as correction keeps pace, and its ceiling is the reciprocal of how far the corrector falls short of one, $\alpha_{\text{crit}} = 1/(1 - \gamma)$. One half is what that law predicts once the independence-accumulation assumption is granted, and it is that assumption the drafted registrations put on trial. The deciding quantity is the ratio of the cross-class to the same-class correction exponent, whose null model, under the assumption that architecture and error structure are irrelevant to correction scaling, is 1.00. Falsification is read in the three-outcome form the programme's drafted registrations use, because refuted-if-the-interval-contains-1.00 would let an underpowered study refute by default: SUPPORTED when the interval on the ratio lies wholly above 1 plus the minimum effect fixed in advance; REFUTED when the interval lies wholly within an equivalence margin around 1.00 fixed in advance (two-one-sided-tests logic), or when any same-class corrector's exponent interval lies wholly above one half, which kills the cap outright; INSUFFICIENT PRECISION when the interval is wider than the margin, reported in exactly those words and never as support or refutation. Both quantities are measurable at current capability on existing systems. The programme's own pilot evidence currently points against the prediction, and the prediction is carried anyway, drafted, preregistered by timestamp (unit 01, first anchored on 28 August 2026 at block 964465) and awaiting human submission, because a prediction recorded against the author's own preliminary evidence is the only kind whose later confirmation means anything.

The objection to pre-empt, because a physicist reaches it in one step. Fault-tolerant quantum error correction suppresses logical error exponentially in code distance below threshold, which looks like a counterexample to any square-root cap. Two scope facts answer it: the exponential suppression runs on the redundancy axis at fixed capability, not on the coupling of corrective strength to the capability of the corrected system, which is the exponent this programme measures; and the threshold theorem itself requires sufficiently uncorrelated physical noise, which is the independence premise again, wearing hardware. Likewise, concentration bounds give exponentially falling error probability while estimator width falls as the square root: different quantities, and conflating them manufactures a refutation.

A second prediction from the same mechanism. The same mechanism yields a second measurable prediction: oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The framework therefore predicts that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.

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Paper XI v1.14 - 5 October 2026: substantive. The title now follows the programme's uniform order (paper, its law or framework, its own title) and reads Paper XI · The ARC Theory · Law I, the ARC Principle · Convergent Evidence for Recursive Amplification: a Graded Cross-Domain Register; until this version it read Convergent Evidence for Recursive Amplification: a Graded Cross-Domain Register Bearing on Law I (Paper XI). The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. §5 and the references now cite the founding paper at version 3.4 of 5 October 2026, the version that carries its uniform title, in which objection A12 is unchanged; the reference entry no longer repeats the paper's designation after its title. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Paper XI v1.13 - 4 October 2026: substantive. The title now names the law the register bears on, Law I, and the heading carries the same title; until this version the paper was titled Convergent Evidence for Recursive Amplification, under the heading Convergent Evidence for Recursive Amplification as a Cross-Domain Structural Principle. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Paper XI v1.12 - 4 October 2026: substantive. A dated note places this paper against Laws IV and V, which were added to the theory on 24 September 2026 and are neither registered as laws nor tested, their mathematics set out at version 1.0 of 3 October 2026. The note says that the paper bears on Law V as context only, and that no result in it is a deciding test of P15, P12 or P21. The epistemic-status line now gives the standing of Laws I to III and of Laws IV and V separately. §5 names Recursive Dynamics as the field the author founded and calls the paper that sets it out the field's founding paper. In §2, Degree 2 now states the PNAS paper's claim in the terms of that paper's own abstract. Four reference entries now print their titles in full as their registries give them (Paper C, Infinite Architects, the PNAS paper, and the first version of the Gumbau paper, whose later versions carry other titles), and the entry for the manuscript of 8 December 2024 says that it is held privately and has no online source. The abstract now describes the PNAS paper and the Gumbau preprint in the terms of their own abstracts. In §2, Degree 2 is headed The Protection Argument, and Degree 3 names the preprint's two theorems and their corollary as its first version does. The draft registrations named in the abstract, §6 and §8 are given as drafted, preregistered by timestamp, with the date of the first anchor, and awaiting human submission. The founding paper is cited at version 3.1 of 3 October 2026. Working labels in §1 and §5 and the word operator in §5 are now plain words. Eighteen works the text cites now have reference entries; six of them are the commentary and the research article of row 5 of the retained snapshot and the four works that its rows 7 to 9 cite. Row 5 links the identifier it prints, that of the commentary, and a bracket names the research article the commentary reports on and gives the day that article's DOI was first registered, 18 November 2025, which is earlier than both dates the row prints; the row's printed words stand. The entry for Pope Leo XIV's encyclical gives the day it was signed, 15 May 2026, beside the day it was presented, 25 May 2026. A second dated note records what was printed before. The works the paper cites are linked to their official sources wherever one was confirmed. The version surfaces and the modification dates move to this revision. Beyond the bracket in row 5 and the links, no row, class, grading or refutation condition changes, and the first published date of 2 July 2026 does not move. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

*Paper XI v1.11 - 21 September 2026: substantive. In the retained pre-grading snapshot, row 4 (the classical time crystal via nonreciprocal forces) now states the external work's first public date, 22 April 2025, the submission date of *arXiv:2504.15495 v1*, and its gap reads 135 days after A1. The fourteen-month figure the row had printed is withdrawn: the canonical evidence register withdrew that gap claim on row EVR-10 on 2 August 2026, and the snapshot row had not been carried with it. Row EVR-10 is corrected at this same revision, from the month-precise 1 April 2025 and 114 days to the day-precise 22 April 2025 and 135 days. The *Physical Review Letters* publication of 6 February 2026 stands as the publication date and the row now states it as such; the row keeps its CONVERGENT class, superseded, as every class in this snapshot is, by the graded register above. As with the other margins this snapshot states against *arXiv* submission dates, first public disclosure has not been separately verified for this row. No other row, class, date, grading or refutation condition changes, the retained snapshot keeps its rows and its superseded label, and the first published date of 2 July 2026 does not move.*

Paper XI v1.10 - 20 September 2026: substantive. The graded register's size and shape are corrected to the register as it now stands, thirty-three rows, generated 6 September 2026 and last amended 7 September 2026, in the abstract, in §1, in the page metadata and in the structured data; §1's class tally is corrected to 1 formal, 2 qualified-technical, 2 engineering, 4 institutional, 6 prior-work, 12 context and 6 excluded; §1 now states that the register discloses four dependence groups, names the two further ones and gives the register's basis for each, and the reading key's count of disclosed dependence groups is corrected with it; the count of source-pending candidate rows is withdrawn, because the published register carries no candidate list; the clause that reader engagement is logged separately is scoped to the rule it states, because no engagement list is published; and the pointer that promised quarantine and engagement lists is corrected to the register itself and to what the evidence spine carries. Six dated notes are attached, one beneath the abstract, one in the reading key, two in §3, one in §6 and one in §7, each in the wording of the owner review of 19 September 2026: they scope the opening claim to what the rows are graded to carry, align the register's evidential role with the tests §6 records as outstanding, class the first two closed-loop properties as intellectual history rather than measured outcomes, apply §1's comparison standard to the third, and carry the defeat threshold's narrowing into the rate language. Two further notes, drafted for §2 and §4, carry a bracketed clause that only the author can settle, whether an

inspection record was kept and whether the search record was kept, and neither publishes at this revision because neither bracket is answered. Editorial. The preregistration banner cites the operational definitions companion at its current version, 1.8.0 of 7 September 2026; the Recursive Dynamics founding paper is cited at v2.8; and the version surfaces and the modification dates move to this revision. No section, row or table is removed. No row, no class, no date, no grading and no refutation condition changes; the retained pre-grading snapshot keeps its rows and its superseded label; and the first published date of 2 July 2026 does not move.

v1.9: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

Paper XI v1.8 - 2 September 2026: the abstract now names two disclosed exceptions, matching the two dependence groups set out in Section 1; in the retained snapshot the Willow row carries both of its dates (preprint 24 August 2024, Nature 9 December 2024), and the FlexHEG row is dated to the month of its arXiv posting; the claim that the three groups were unaware of one another is corrected, since the Gumbau paper builds on Hernández-Espinosa's; the PNAS paper's argument reads as one about machine agents, with the human parallel attributed to Paper C; the Gumbau theorems carry a date and are described as unrefuted rather than proved; the defeat-threshold paragraph is reconciled with the no-total policy; the ceiling reads as the reciprocal of the shortfall, $\alpha_{crit} = 1/(1 - \gamma)$, replacing the retracted reciprocal-of-the-exponent form the text had printed; the Spotlight status of Engels et al. is verified; and the references are corrected (Greenblatt et al. for the alignment-faking paper, both Willow dates, Paper X's current title). v1.7 - 1 September 2026: dating convention and register alignment.

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BIBTEX

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convergence.html}  
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Eastwood, 2026

See /how-to-cite/ for the canonical citation manual across all artefacts, or /citations.json for the machine-readable index.

Epistemic status. What this programme names Laws are conjectures. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026; the predictions that carry them are registered, the laws themselves are not yet registered, and neither has been tested. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

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Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

The ARC Theory · ARC/Eden experiments · Paper XI · v1.14 · 2 July 2026, revised 5 October 2026

OSF: [10.17605/OSF.IO/DC9GW](https://doi.org/10.17605/OSF.IO/DC9GW) · michaeldariuseastwood.com/research/evidence-spine.html

The method IS the hypothesis. A dated, graded convergence record, every row carrying its primary source and its evidence class. The closed loop is documented end to end.

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